

Tide gauge survey of the outer harbor

Sample document written for a parsing test. All stations and readings are invented.

Four gauges along the outer harbor wall were read every ten minutes for thirteen weeks. This report gives the mean water level at each station, the largest departure from the predicted tide, and the share of readings that each gauge delivered. One gauge was replaced in week six, and its two instruments are reported together.

1 Method

Each reading is the level above chart datum, in centimeters. The residual of a reading is the measured level minus the predicted tide at the same minute:

$$r(t) = h(t) - p(t)$$

- A reading is dropped when the gauge reports a fault code.
- A week counts as complete when at least 95% of its readings are kept.
- Station D is reported from week six, after its sensor was replaced.¹

2 Results

Mean levels differ by less than four centimeters between stations. The largest residual was recorded at station B during the storm of week nine.

Table 1. Water level, largest residual and kept readings per station.

Station	Water level (cm)		Largest residual (cm)	Readings kept
	Mean	Standard deviation		
A, north mole	212.4	61.8	+38	99.1%
B, fairway	214.9	63.2	+57	98.4%
C, lock gate	211.2	60.5	+41	97.7%
D, south quay	213.6	62.0	+44	96.2%

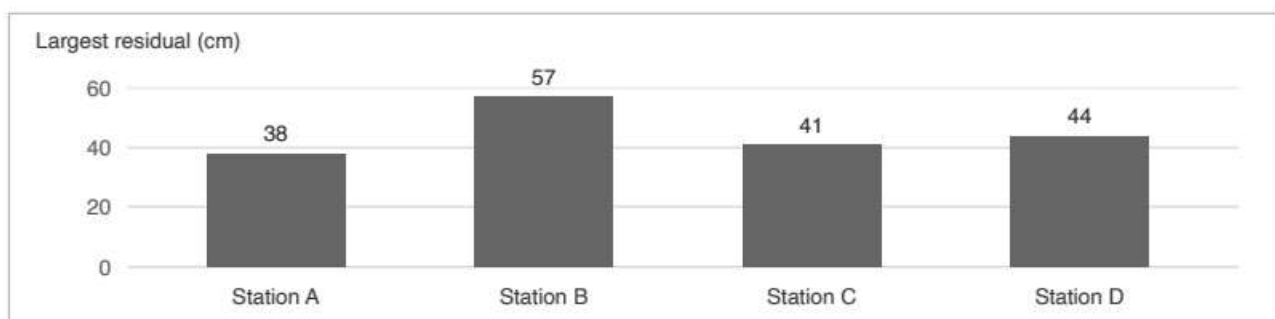


Figure 1. Largest residual at each station over the quarter.

¹ The replaced sensor at station D was read against the old one for two days before the old one was removed.

2.1 Weekly record

The table below gives the mean level of every week at every station. A week is the seven days that begin on Monday at midnight, local time. Weeks with fewer than 95% of their readings are printed like the others and are listed in the discussion.

Station D has no value before week six. Its first sensor drifted by more than two centimeters a day, and readings from that sensor were not used for any figure in this report.

Table 2. Mean water level of each week at each station, in centimeters.

Week	Station A	Station B	Station C	Station D
1	211.8	214.2	210.6	not read
2	212.1	214.6	210.9	not read
3	213.0	215.4	211.7	not read
4	212.6	215.1	211.3	not read
5	211.5	213.9	210.2	not read
6	212.2	214.7	211.0	213.1
7	212.9	215.3	211.6	213.8
8	212.0	214.5	210.8	213.2

Table 2 (continued).

Week	Station A	Station B	Station C	Station D
9	214.3	217.8	212.9	215.6
10	212.7	215.0	211.4	213.9
11	212.3	214.8	211.1	213.5
12	211.9	214.4	210.7	213.0
13	212.5	214.9	211.2	213.4

Week nine stands out at every station. A storm from the northwest held the water in the harbor for two days, and the mean of that week is about two centimeters above the mean of the quarter.

3 Discussion

The four stations agree on the shape of the quarter. The differences between them are steady from week to week, which is what a fixed offset in the height of each gauge would produce, and they are small against the tide itself.

3.1 Station D

The new sensor at station D was compared with a staff reading on three mornings. The largest difference was four millimeters. Two things remain open:

- The old and the new sensor were mounted eleven centimeters apart, and no correction was applied for it.
- Week six holds 91% of its readings at station D and does not count as complete.

4 Conclusion

All four gauges delivered more than 96% of their readings over the quarter. The survey continues in the fourth quarter with the same stations and the same method.

